F 03-09

sanas South African National Accreditation System

NON-CONFORMANCE, CORRECTIVE ACTION AND CLEARANCE REPORT

SANAS No.: T1195

Organisation Name: AZURESKY PTY (LTD)

NC Report Number: NC04

Area/Scope Assessed: Coal Testing

Assessor / Inspector / Expert: NCHILIZA

PART 1: DETAILED OBSERVATION

The Control Chart for Volatile Matter (27/08/25 – 3/10/25) had ten Consecutive points below the mean and no action was taken

1) The rules presented for Control Chart does not form part of the System as they are not dated or referenced form the Lab Procedure (Ensuring validity of Results

Signed (Representative):

Signed (Team Leader / Inspector):

MAJOR: [X] MINOR: [ ]

Reference No. of relevant Guide / Standard / Requirement: ISO/IEC 17025 7.7

Date: 08/04/26

PART 2: ROOT CAUSE ANALYSIS

(Provide an analysis of the extent and cause of the nonconformity)

The QMS lacked defined processes on how control charts are interpreted and evaluated. In addition, the bias on the VM control chart was not investigated due to negligence.

PART 3: CORRECTIVE ACTION TAKEN

(Clearly indicate what corrective action was taken, and attach supporting evidence of implementation)

The AS001V procedure has been updated to include how control charts are interpreted and to also reference NC/work procedure in addressing deviations. Staff was also informed on the changes of the procedure (See register).

Date: 13/04/2026

Signed (Representative):

PART 4: CORRECTIVE ACTION VERIFIED AND CLEARED

(Comment on whether the corrective action is sufficient and appropriate)

Date:

Signed: Assessor / Inspector/Expert:

PART 5: FOLLOW-UP AT NEXT ASSESSMENT

(Comment on the implementation of the corrective actions)

Date:

Signed: Assessor / Inspector/Expert:

©SANAS

AS004MF 07/25

BRAINSTORMING FORM

IRQ no: NC04 (SANAS)

Brainstorming team

Ifiso Nyambi
Nosipho Mlonyeni
Abenathi Tana.
Tshepo Maponya.

What went wrong? Control chart for VM (27/08/25 - 3/10/25) had 10 consecutive points below the mean and no action was taken. In addition; the rules presented does not form part of the system as they are not dated or referenced from the lab procedure.

Why 1: Why the VM control chart with bias not investigated? Although training was provided on the investigation of bias on control charts, there was no documentation on the QMS on the interpretation of control charts and how the deviations will be addressed.

Why 2: Why there was no proper documentation? The processes not defined and/or incomplete system documentation.

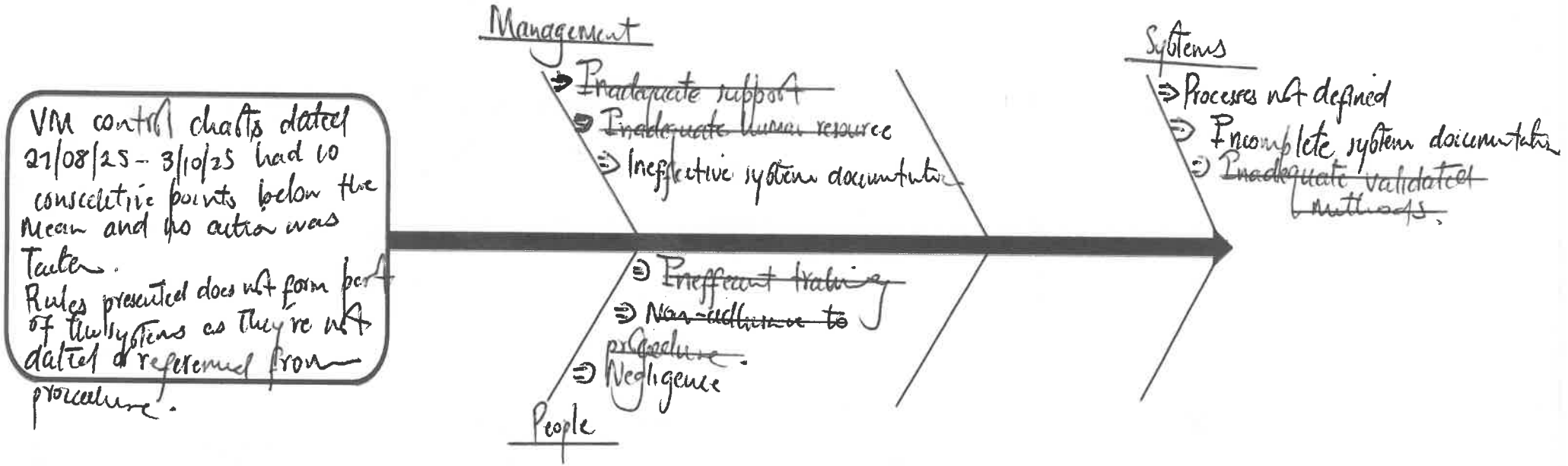
Supervisor:

AS001MF

07/25

ROOT CAUSE ANALYSIS FORM

IRQ NO: NCO4 (SANAS)



SUPERVISOR

AzurESky

ATTENDANCE REGISTER

Document no:	Version:	Page:
ASMF 17043-8.4.1	0	1
Compiled by:	Sifiso Nyambi - Laboratory Manager	
Approved by:	Nonhlanhla Msibi - Operations Manager	

Date:	13/04/2026		
Chair:	Sifiso M	Scribe	
Subject:	Control Charts interpretation and bias investigation (Update on AS001V).		
Absent:			

Present	Cell Number	Designation	Signature	Email address
N.N Hlongwane	0723732558	Lab Analyst		nosipho@Azuresky.it.com.
A Jama	0606111584	Lab analyst		abenathi.jama@gmail.com
TMJ Maponya	076 455 0692	Lab Supervisor		tshepo@azuresky.it.com
SB Nyambi	072174 1852	Lab Manager		sifiso@azuresky.it.com.
N. Msibi	0809368149	Ops. Manager		nonhlanhla@azuresky.it.com
"	"	"	"	

Before

- The measured value is compared with the certified value; deviations beyond control limits (reproducibility) are investigated.

3. **Control Charts:**
   - Control charts are maintained for each parameter using CRM and duplicate data.
   - Each new result is plotted and reviewed to detect bias, drift, or loss of precision.
   - Out-of-control points (beyond Reproducibility limits) or systematic trends require immediate suspension of affected runs until resolution.

5.3 Method Validation and Ongoing Verification

1. All methods are **validated before routine use** to confirm accuracy, precision, linearity, and detection limits.
2. Periodic re-validation is performed after any significant change (New equipment).
3. Analysts verify method performance using QC data trends and inter-analyst comparisons.

5.4 Proficiency Testing and Interlaboratory Comparison

1. The laboratory participates in **proficiency testing (PT)** for each accredited parameter.
2. PT performance is evaluated using **z-scores** and included in management review.
3. If no PT is available, **inter-laboratory comparisons** are conducted

---

5.5 Trending and Review

Effective date	Rev status	Next review date
10/2025	00	10/2028

CONTROLLED DOC

Before

	AzureSky		
AzurESky "ensure quality of test"	Ensuring Validity of Results (Clause 7.7)		Document no: AS001V
	Compiled by:	Tshepo Maponya – Laboratory Supervisor	
	Approved by:	Sifiso Nyambi – Laboratory Manager	

Record	Retention Period	Responsibility
Duplicate Test Records	3 years	Analyst
PT/Inter-lab Results	5 years	Operations Manager
IRQs and Corrective Actions	5 years	Quality Manager

8. Review

This procedure shall be reviewed annually during management review meetings.

9. References

- ISO/IEC 17025:2017 — Clauses 7.6, 7.7, 7.8
- ILAC G8:09/2019 — Guidelines on decision rules and reporting
- ASTM D3173, D3174, D3175, D5865 — Coal analysis standards
- Laboratory QMS Manual, Section 7 — Quality Control and Validation

10. Revision

Revision no	Content changed	Reason for revision	Date changed

Effective date	Rev status	Next review date
10/2025	00	10/2028

	Azuresky		
AzurESky ensure quality of test	Ensuring Validity of Results (Clause 7.7)		Document no: AS001V
	Compiled by:	Tshepo Maponya – Laboratory Supervisor	
	Approved by:	Sifiso Nyambi – Laboratory Manager	

2. **Certified Reference Material (CRM):**
    - A CRM is analysed **after every 10 routine runs** for each parameter.
    - The measured value is compared with the certified value; deviations beyond control limits (reproducibility) are investigated.
3. **Control Charts:**
    - Control charts are maintained for each parameter using CRM and duplicate data.
    - Each new result is plotted and reviewed to detect bias, drift, or loss of precision.
    - Out-of-control points (beyond Reproducibility limits) or systematic trends require immediate suspension of affected runs until resolution.

Interpretation of control charts and bias:

- 8 points on one side of the mean constitutes a bias.
- The lab personnel shall start an investigation as soon as 5 points are observed on one side of the mean, a letter shall be placed next to the 5th point to indicate action. There need not be records for the 5 points. This is to say deviations will only be noted down when the 8 points are observed.
- The lab shall also pay attention to points falling outside the upper/lower limits. 3 consecutive points outside these limits shall trigger an investigation.
- The lab shall follow the NC work procedure in addressing deviations under this section.

Effective date	Rev status	Next review date
04/2026	01	04/2029

AFTER

	Azuresky	
	Ensuring Validity of Results (Clause 7.7)	**Document no: AS001V**
Compiled by:	**Tshepo Maponya – Laboratory Supervisor**	
Approved by:	**Sifiso Nyambi – Laboratory Manager**	

Record	**Retention Period**	**Responsibility**
IRQs and Corrective Actions	5 years	Lab Manager/Supervisor

8. Review

This procedure shall be reviewed annually during management review meetings.

9. References

- ISO/IEC 17025:2017 — Clauses 7.6, 7.7, 7.8
- ILAC G8:09/2019 — Guidelines on decision rules and reporting
- ASTM D3173, D3174, D3175, D5865 — Coal analysis standards
- Laboratory QMS Manual, Section 7 — Quality Control and Validation

10. Revision

Revision no	Content changed	Reason for revision	Date changed
00	Added PSD and TM under section 2	Lab will want to add these two parameters in the future for the extension of scope.	13/04/2026
01	Add section for the interpretation of control charts and how the deviations shall be addressed	There was a gap in the system with regards to how control charts are assessed, as well as how deviations from the charts will be addressed.	13/04/2026

Effective date	Rev status	Next review date
04/2026	01	04/2029